Cross Validation

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Goal

- Reduce the bias of the estimated empirical generalization error

Cross Validation

Definition

- Take different subsets of (training) data
- Train one model on each subset and test on the remaining disjoint subset of (training) data
- Average the test performance

Cross Validation

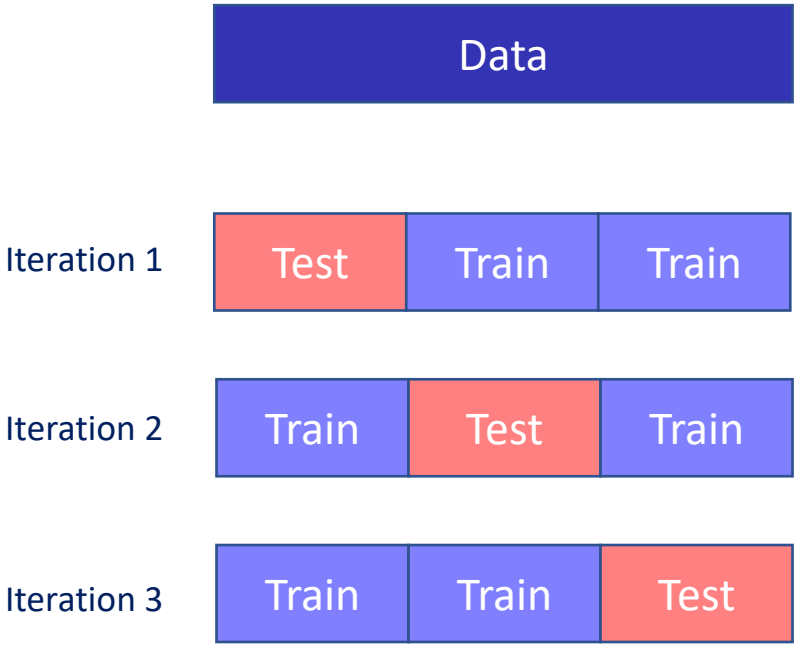
Goods and Bads

- Reduced bias in our estimate of the generalization error
- Efficient use of data
- You can make sure that all samples end up in the test set at least once
- It’s computationally expensive

Cross Validation

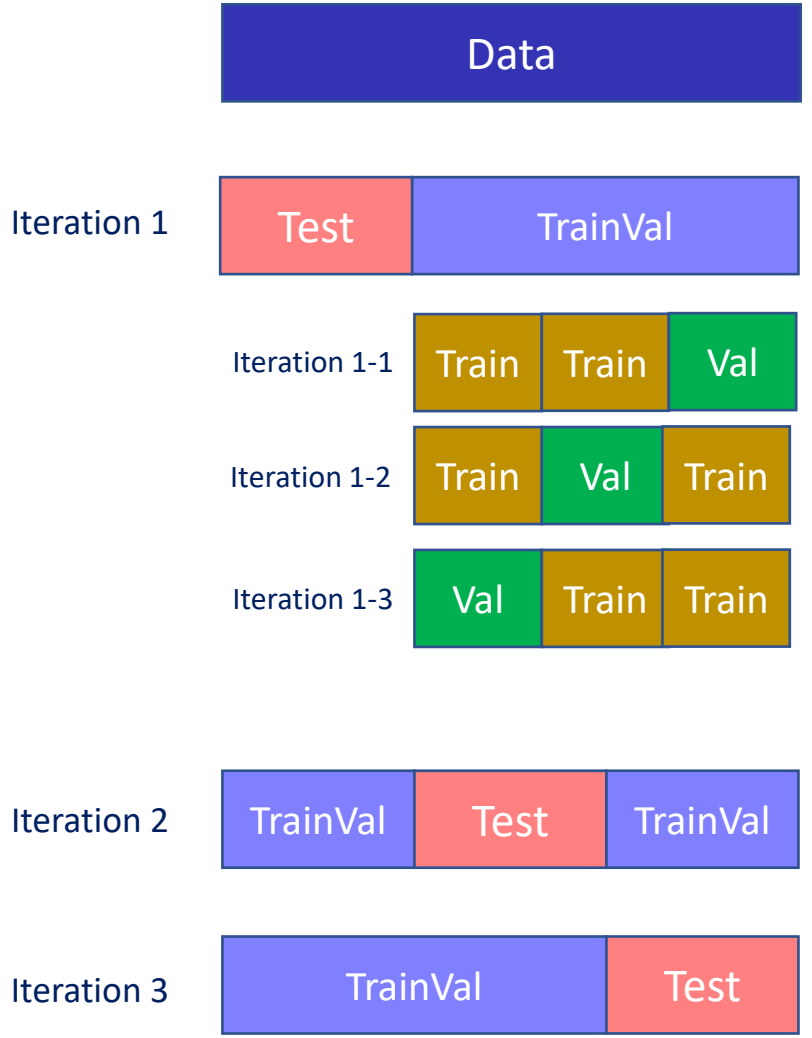
K-fold Cross Validation

- Divide the (training) data into k equally-sized folds
- Train K models, each one by taking the k'th fold as test set and the rest as training set



Nested Cross Validation

Ideally, we want to cross validate for both HP search and test error



High-Computational-Budget Regimes

- Should I increase K (number of folds) to approach the size of dataset?
- Surprisingly no!
- e.g., in the case of data violating i.i.d. assumption, high-K might accentuate the underestimation of the generalization error.

Notes

- Depending on the evaluation metric, to get cross-validation accuracy/error, you cannot always combine the scores/errors of samples across different folds to do one evaluation (e.g. AUROC)
  - better to combine the evaluations across subsets (e.g. mean AUROC)
  - but accuracy can be combined across different subsets (same as mean)
- Don’t use data at all before cross validation
  - common in genomics → e.g. feature selection before the modeling

Reflection/Discussion point

- Should we (always) randomly split the data to train/val/test or into folds for cross validation?



Try to think about your trained model in use, to what kind of samples do you want it to generalize?

3 minutes

Principle of maximal independence

- In general we would like our test subset to be able to capture the "generalization" error, so train, validation and test sets should reflect different aspects of generalization:
  - over samples (individual patients in different subsets)
  - over time (different times in different subsets)
  - over geographical locations (different countries in different subsets)
  - …

Can we trust these “mean” values though?